

UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have made substantial progress across the broad landscape of visual question answering, encompassing tasks as diverse as visual reasoning, document understanding, and multimodal dialogue. These gains hold widely across VLMs built on a variety of base language models, alignment architectures, and training corpora, suggesting a general trend rather than an isolated one. Yet a growing body of work indicates that, across this same family of models, performance lags on traditional image classification benchmarks that probe fine-grained visual knowledge. We evaluate a large and diverse set of recent VLMs on fine-grained classification benchmarks spanning multiple visual domains, and we identify factors that generalize across this disconnect between fine-grained knowledge and other vision benchmarks. Through a broad series of ablation experiments that vary language models, vision encoders, and training paradigms, we find that a stronger LLM tends to raise all benchmark scores uniformly, whereas a stronger vision encoder disproportionately improves fine-grained classification across configurations. We further find that the pretraining stage is consistently vital to fine-grained performance, particularly when the language model weights are unfrozen during pretraining. Taken together, these insights point toward general strategies for enhancing fine-grained visual understanding and, more broadly, the vision-centric capabilities of VLMs.

1 INTRODUCTION

Recent advances in vision-language models (VLMs), which integrate vision encoders with large language models (LLMs), have demonstrated increasingly sophisticated capabilities in interpreting and reasoning about visual content across a wide array of settings (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). Spanning many architectures and training regimes, these models can perform complex tasks ranging from open-ended visual question answering to document understanding and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Despite this breadth of progress, a crucial and broadly applicable question remains: How well do VLMs, as a class, perform on fine-grained visual perception tasks, and how can we improve their performance in a way that generalizes across models?

Fine-grained visual recognition—the ability to distinguish between visually similar categories, a challenge that recurs across essentially every visual domain—requires models to focus on subtle distinguishing features while maintaining robustness to the many forms of intra-class variation encountered in the wild, such as changes in pose, lighting, and background. While traditional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have demonstrated strong performance across a range of fine-grained classification benchmarks, the corresponding behavior of the newer and rapidly growing family of VLMs has received comparatively little attention.

Understanding the fine-grained knowledge capabilities of VLMs is essential because perception serves as the foundation for a wide range of advanced capabilities, including understanding and reasoning. Across the many settings in which they are deployed, real-world applications of VLMs frequently involve fine-grained, long-tailed distributions where distinguishing between closely related categories is crucial. For instance, if a model fails to identify a mushroom species correctly, it cannot determine whether it is poisonous based on textual knowledge or whether it is safe for consumption—and analogous failure modes arise across domains from wildlife identification to re-

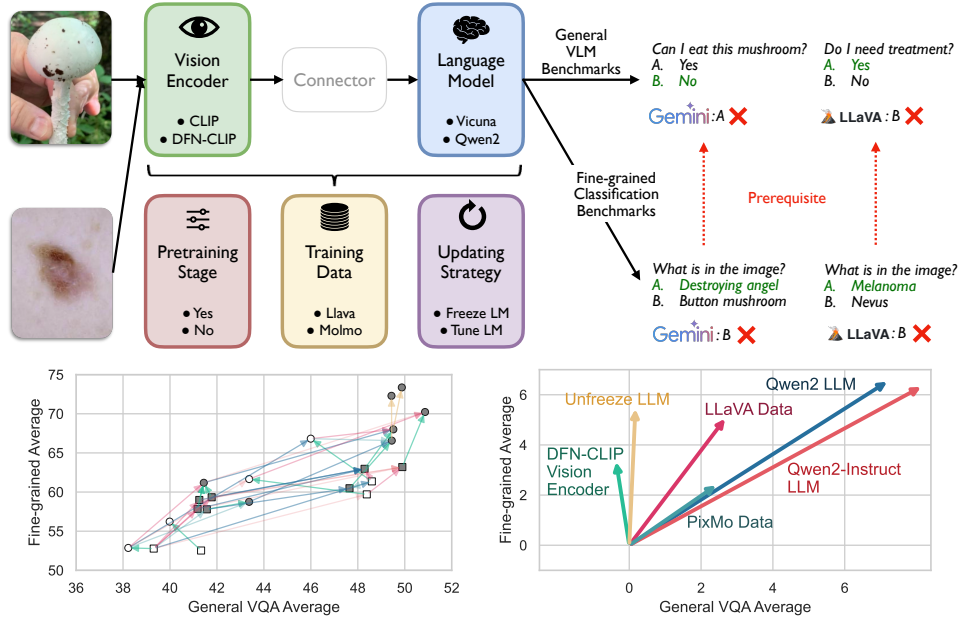


Figure 1: **Overview.** (Top) We investigate the fine-grained classification capabilities of vision-language models (VLMs) across diverse visual domains, a crucial yet often overlooked aspect that broadly underpins higher-level understanding and reasoning. (Bottom) Through 22 systematic ablation experiments spanning key model components and training strategies (left), we identify factors that consistently drive fine-grained classification performance in VLMs (right).

tail and industrial inspection. Moreover, insights into the factors that influence fine-grained knowledge acquisition are likely to inform more effective architectures and training strategies for future VLMs in general, beyond any single model or dataset.

In this paper, we first present a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs, comparing their performance on fine-grained classification benchmarks with their capabilities on general VLM tasks. To capture a representative cross-section of the field, we evaluate 15 VLMs spanning distinct base models, architectures, and training pipelines—including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024)—on four fine-grained classification benchmarks drawn from different visual domains, namely ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014), alongside aggregated results from eight general VLM benchmarks such as MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). Across this diverse pool, our analysis reveals significant variance in fine-grained classification performance among models with comparable general abilities, suggesting that fine-grained recognition represents a distinct axis of visual intelligence that current VLM benchmarks broadly fail to capture. Furthermore, we observe a persistent gap between VLMs and their CLIP-based vision encoders that recurs across models, mirroring and generalizing previous findings (Zhang et al., 2024; Geigle et al., 2024).

To better understand what contributes to fine-grained classification performance in VLMs, we conduct a systematic study examining the role of key model components and training paradigms that are common across the VLM landscape. Specifically, we investigate whether widely shared VLM components, such as language models and vision encoders, impact classification performance, and whether commonly used training strategies, such as pretraining or variations in data quality, influence results. Through 22 carefully designed ablation experiments that vary these components and strategies, we identify several factors that generalize across settings and significantly affect fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Improvements in the base language model tend to enhance performance uniformly across both fine-grained classification and general VLM benchmarks, a pattern that holds across the configurations we study.

- **Vision encoder (§4.2):** A stronger vision encoder consistently improves fine-grained classification performance but has a limited impact on general VLM benchmarks, particularly for models trained in a two-stage fashion.
- **Pretraining stage (§4.3):** Large-scale pretraining on image-captioning datasets broadly boosts fine-grained classification performance while having a lesser effect on general VLM benchmarks.
- **Weight updating method (§4.3):** Fine-grained classification performance improves across settings when both the connector between the vision encoder and LLM and the LLM itself are trained during pretraining, rather than training only the connector.
- **Data quality (§4.3):** The quality of pretraining and instruction-tuning data has a limited impact on fine-grained classification in general, particularly when language model weights are frozen during pretraining.

In summary, our study examines fine-grained visual perception in VLMs—an essential yet broadly overlooked aspect of current VLM benchmark evaluations. Through systematic ablations spanning a range of model components and training choices, we identify factors that generalize across settings and govern fine-grained knowledge acquisition in VLMs. These findings offer broadly applicable insights for developing vision-centric VLMs with enhanced visual understanding, improving their effectiveness across the many real-world long-tail tasks that demand fine-grained recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) constitute a broad class of models that integrate visual inputs with language models, most commonly following the LLaVA-based architecture (Liu et al., 2023) built from three core components: a vision encoder, a language model, and an MLP connector bridging the two. Across this family, these models have achieved strong performance on a wide variety of multimodal tasks, including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). To evaluate them, a large and growing collection of benchmarks has been developed, such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). However, these benchmarks primarily assess reasoning and language understanding given visual inputs, while broadly overlooking core vision-centric capabilities such as object recognition and fine-grained classification. To bridge this gap, in this work we systematically evaluate and analyze VLMs on fine-grained classification, an essential yet underexplored dimension of visual intelligence that cuts across models and domains.

Fine-grained visual classification. Fine-grained classification is a well-established computer vision task that aims to distinguish between visually similar subcategories within broader object categories, and it spans a wide range of domains, from bird species (Wah et al., 2011) and flowers (Nilsback & Zisserman, 2008) to pet breeds (Parkhi et al., 2012) and beyond. Benchmarks of this kind have played a key role in training and evaluating a broad succession of vision encoders, from ResNet (He et al., 2016) to CLIP (Radford et al., 2021) and DINO (Caron et al., 2021). Despite their long-standing importance in vision research, recent VLMs have largely overlooked these datasets in their evaluation protocols in favor of visual question-answering. Because fine-grained classification underlies a wide spectrum of real-world applications that involve long-tail distributions and demand precise recognition—for example, medical diagnosis, food safety, and species identification, among many others—evaluating VLMs on such benchmarks is essential for understanding their practical usability from a vision-centric perspective, beyond language-centric reasoning tasks.

Evaluation and ablation of VLMs for fine-grained classification. While recent studies have begun to explore VLM evaluation on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), they have generally been limited in scope, often focusing on just a few models or a single dataset. We broaden these efforts by evaluating a much wider range of VLMs across multiple fine-grained classification benchmarks spanning distinct visual domains, all cast in a unified multiple-choice format that enables consistent comparison. Building on this broader evaluation, we go beyond prior work with a comprehensive ablation analysis aimed at identifying factors that generalize across models and settings to drive fine-grained classification performance.

Several recent works have investigated the design space of VLMs, analyzing aspects such as model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of

multimodal datasets (Gadre et al., 2023; Udandara et al., 2024; Deitke et al., 2024). However, these studies have largely centered on general benchmarks that are more language- and reasoning-focused. In this work, we instead investigate how key design choices that are shared broadly across VLMs—including base model selection, pretraining methods, and the composition of pretraining and instruction-tuning data—affect fine-grained classification, providing insights that we expect to transfer to optimizing VLMs for vision-centric tasks more generally.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

In this section, we discuss the motivation for evaluating fine-grained classification, describe our benchmarks and evaluation setup, and present our key observations from testing 15 existing VLMs that together span a broad range of base models, architectures, and training pipelines.

3.1 MOTIVATION

Fine-grained classification is a fundamental task in computer vision that focuses on distinguishing visually similar subcategories within a broader category, a challenge that arises across many domains (e.g., different bird species, car models, or plant varieties) (Wei et al., 2021). A wide range of evaluation benchmarks, such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008), have been developed and widely adopted as standard testbeds for training and evaluating vision encoders. With the advent of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have achieved near- or superhuman performance on challenging fine-grained classification tasks across such benchmarks.

However, this broad class of benchmarks has been largely overlooked in the evaluation of vision-language models (VLMs). While recent VLMs like LLaVA (Liu et al., 2023) have demonstrated strong capabilities in interpreting visual concepts, their evaluation is typically confined to visual question answering (VQA), including academic VQA benchmarks like MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding tasks like DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). These benchmarks emphasize reasoning and language processing but lack fine-grained object categories, thereby leaving a core, broadly relevant vision-centric capability largely unexamined.

Fine-grained classification is crucial for VLMs, as the many real-world settings in which they are deployed often present long-tailed distributions that require precise visual knowledge to separate similar categories. For example, in daily life, if a blind person asks whether a mushroom is safe to eat, the model must first correctly identify the species before reasoning about its toxicity—misclassifying a deadly destroying angel as an edible white button mushroom could have fatal consequences. Similarly, in medical diagnosis, failing to distinguish between similar diseases could lead to incorrect treatment and serious health risks. In self-driving cars, confusing a “stop” sign with a “do not enter” sign—both red with similar shapes—could result in dangerous navigation errors. These examples are drawn from very different application domains, yet they share the same underlying demand: it is critical for VLMs to excel at fine-grained classification to ensure safety and reliability across the full breadth of real-world scenarios.

3.2 BENCHMARKS

We evaluate the fine-grained visual knowledge of VLMs using four well-established object recognition benchmarks that were deliberately chosen to cover a diverse spread of visual domains—from general objects to flowers, animals, and food: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014). This diversity lets us probe fine-grained capability broadly rather than within any single category.

These benchmarks span from 37 to 1,000 distinct classes, a wide range that makes direct evaluation of VLMs in a multiple-choice format—the format commonly used in VLM training and evaluation—challenging. To handle this range uniformly, we adopt the methodology of Geigle et al. (2024) to convert each dataset into 5-way multiple-choice questions. Specifically, for each image, we use an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) to select the four hardest negative

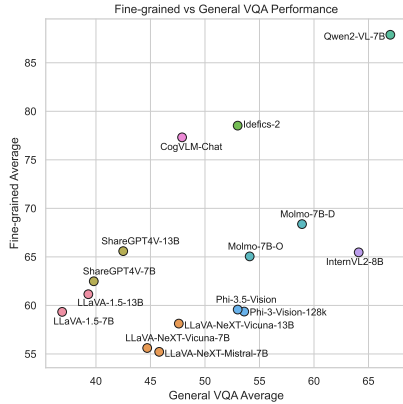


Figure 2: **Fine-grained classification compared to general VQA performance across VLMs.** Across a diverse set of recent VLMs, fine-grained classification emerges as a distinct aspect of visual capability that standard VQA benchmarks broadly fail to measure.

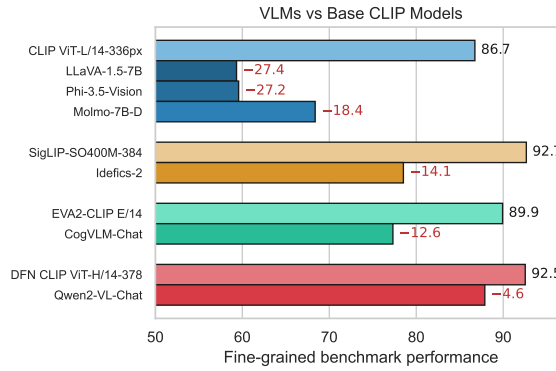


Figure 3: **Comparison of VLMs with their corresponding CLIP vision encoders in fine-grained classification.** While Qwen2-VL-Chat nearly matches the performance of its vision encoder DFN-CLIP, the remaining VLMs consistently fall well behind. This pattern, recurring across models, highlights that VLMs in general have considerable room for improvement on fine-grained classification tasks.

choices based on image-label cosine similarity and combine them with the ground-truth label to form the multiple-choice answers. We deliberately choose the OpenCLIP model so that difficulty is not biased against any particular family of VLMs, since these rely on a variety of encoders such as OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023) (all trained on different datasets than the OpenCLIP model). This conversion to multiple choice lets us test a broad range of VLMs in their native format while preserving much of the underlying difficulty. For instance, CLIP ViT-L/14-336px (Radford et al., 2021) obtains 75.2% accuracy on the 102-class Flowers-102, which rises only modestly to 78.3% on our 5-way multiple-choice version.

3.3 TESTING SETUP

We evaluate 15 recent VLMs spanning the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abdin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b); this pool was chosen to cover a diverse cross-section of publicly available architectures and training recipes. For each VLM, we use its respective default prompt format from VLMEvalKit (Duan et al., 2024) and measure accuracy based on the exact match with the multiple-choice options. For baseline comparison, we evaluate a range of CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) in the standard zero-shot setup, where both images and class labels (“A photo of a [class]”) are encoded using CLIP’s image and text encoders. The top-1 prediction is determined by computing the cosine similarity between image and text embeddings.

To assess how broadly fine-grained classification capabilities are captured by existing VLM evaluation benchmarks, we compare VLM performance on the fine-grained classification benchmarks against their average score across eight general VQA benchmarks, such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), from VLMEvalKit.

3.4 RESULTS

Figure 2 shows the relationship between fine-grained classification performance (averaged across our four benchmarks) and general VQA performance (averaged across eight existing VQA benchmarks) for all tested VLMs. Across this diverse set of models, we observe that VLMs with similar performance on general VQA benchmarks nonetheless exhibit substantial variation in fine-grained classification accuracy. For example, while CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both achieve nearly 48% average accuracy on general VQA bench-

marks, their fine-grained classification accuracy differs by 19pp (percentage points)—77.3% for CogVLM-Chat and 58.1% for LLaVA-NeXT-Vicuna-13B. This large discrepancy, which recurs throughout the pool rather than being confined to a single pair of models, indicates that fine-grained classification capabilities are broadly not captured by existing VLM evaluation benchmarks, marking fine-grained classification as a distinct dimension of visual intelligence.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks broadly fail to measure adequately.

A general property shared by current VLMs is that each is built on top of an existing pretrained vision encoder. Because of this, the multiple-choice format lets us directly compare performance between VLMs and their underlying CLIP models on traditional classification benchmarks, across the full range of models we consider. Figure 3 compares the fine-grained classification performance of selected VLMs with their corresponding CLIP model baselines, revealing a substantial gap that persists across models. For instance, Molmo (Deitke et al., 2024), a very recent model that excels on a variety of VQA benchmarks, obtains only 68.4% average accuracy in our multiple-choice evaluation setting, while its vision encoder CLIP ViT-L/14 (Radford et al., 2021) obtains 86.7%. Even the best tested model, Qwen2-VL-Chat (Wang et al., 2024b), falls 4.6 points behind its own state-of-the-art DFN-CLIP (Fang et al., 2023) vision encoder. The consistency of this performance gap across otherwise very different models underscores the need to analyze design choices in VLMs broadly, so as to identify factors that enhance fine-grained classification capabilities in general. In the next section, we conduct an ablation study to investigate these factors.

Finding 2: A significant performance gap between VLMs and their CLIP vision encoders persists broadly across models on fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on our observational results in Section 3, we ablate the key differences between models that may contribute to fine-grained classification performance, focusing primarily on base model selection and training methodology—two axes that vary widely across the VLM landscape. As a controlled and representative testbed, we build on the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023), which is broadly representative of the dominant VLM design. All models undergo one epoch of pretraining (when applicable) followed by one epoch of finetuning. We use LLaVA-1.5’s default hyperparameters, including learning rate and batch size, across all experiments so that observed effects can be attributed to the components being varied.

Fine-grained performance is evaluated as the average score across the four classification benchmarks spanning distinct visual domains described in Section 3.2. For general VQA performance, we report the average score across three widely-used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These benchmarks were selected for their broad domain coverage and widespread adoption in VLM evaluation.

4.2 BASE MODELS

Our analysis in Section 3 indicates that, across recent models, newer base LLMs are widely adopted, whereas the CLIP ViT-L/14 vision encoder from Radford et al. (2021) remains a near-universal choice. We investigate how these two base model choices—both common across the field—impact fine-grained classification performance.

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as our baseline, we ablate the LLM component across a range of vision encoders and pretraining configurations, so that any effect we observe reflects the LLM broadly rather than a single setting. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) (used in LLaVA) against Qwen2-7B (Yang et al., 2024) (used in Qwen2-VL-7B (Wang et al.,

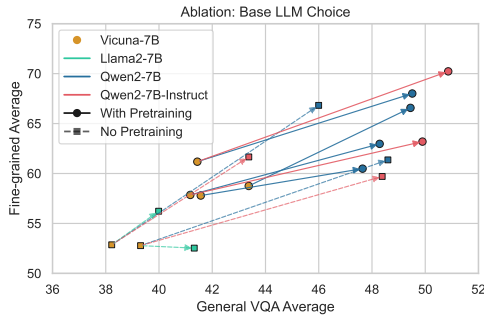


Figure 4: **Ablating base LLM, swapping out Vicuna-7B from LLaVA with other LLMs.** Averaged across configurations, switching from Vicuna to Qwen2-7B yields a +7.5pp increase in fine-grained performance and a +7.7pp improvement in general VQA performance, a benefit that holds across the settings tested.

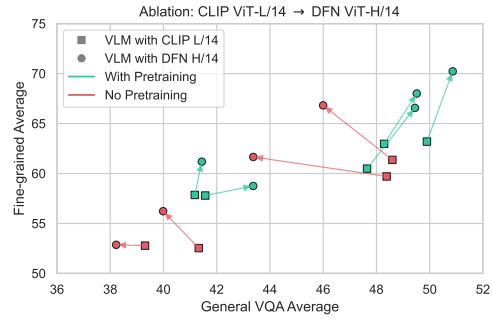


Figure 5: **Ablating base vision encoder, swapping out CLIP ViT-L/14 from LLaVA for DFN-CLIP ViT-H/14.** We find that, given sufficient pretraining, switching to DFN-CLIP improves fine-grained performance by +4.5pp and general performance by +1.2pp, indicating a broadly beneficial effect of stronger encoders.

2024b) and Molmo-7B-D (Deitke et al., 2024)), as well as Llama-2-7B (Touvron et al., 2023) (Vicuna’s base model) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B), together spanning both instruction-tuned and non-instruction-tuned models.

Findings. As shown in Figure 4, replacing Vicuna-7B with Qwen2-7B substantially improves performance across every configuration we test, with average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. The instruction-tuned Qwen2-7B-Instruct shows comparable improvements (+7.5pp and +8.1pp, respectively). Llama2-7B, the non-instruction-tuned base of Vicuna, yields more modest improvements, though the downstream implications of using a model without instruction tuning remain unclear. Notably, all of our trained models—including the lower-performing Vicuna-based VLMs—consistently produce multiple-choice responses in the correct format. This uniformity suggests that performance differences in fine-grained classification stem from improved knowledge about the presented options rather than better format adherence, an interpretation that holds across the range of LLMs considered.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks, a trend that generalizes across the configurations we examine.

4.2.2 VISION ENCODER CHOICE

Setup. Across the VLMs we evaluate, the OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) encoder is by far the most common choice, used by the LLaVA series, Molmo, and many others. A notable exception is Qwen2-VL (Wang et al., 2024b), which uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), an encoder that demonstrates superior zero-shot performance on object recognition tasks such as ImageNet. We evaluate both vision encoders across a range of LLM choices and pretraining configurations to characterize their effect broadly.

Findings. Figure 5 illustrates that the impact of substituting CLIP L/14 with DFN-CLIP H/14 depends on whether connector pretraining is performed before finetuning. Without pretraining, DFN-CLIP degrades general VQA performance (-2.5pp) while modestly improving fine-grained benchmark scores (+2.8pp). However, once models have undergone connector pretraining, general VQA performance remains stable or slightly improves (+1.2pp) while fine-grained performance increases substantially (+4.5pp). This indicates that, given adequate integration, stronger vision encoders can broadly boost fine-grained classification performance in pretrained VLMs.

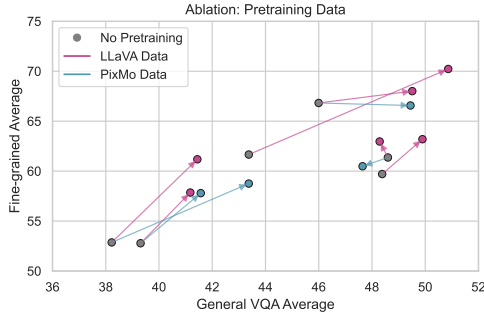


Figure 6: **Ablating pretraining data.** Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) data yields broadly similar gains regardless of the caption source. LLaVA data increases fine-grained and general scores by +4.0pp and +2.1pp respectively, while PixMo data increases them by +2.4pp and +2.5pp.

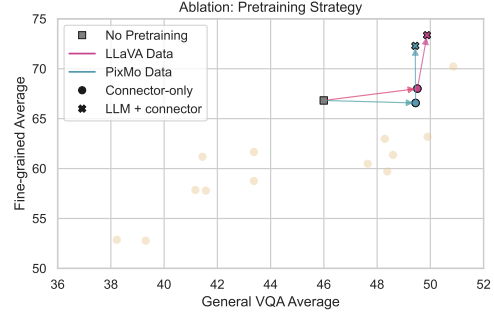


Figure 7: **Ablating training method, comparing connector-only pretraining with LLM+connector tuning.** Unfreezing the LLM during pretraining yields a +5.5pp improvement in fine-grained performance without adversely affecting general VQA scores, an effect that holds across the pretraining data sources considered.

Table 1: **Summary of important ablation settings and comparison to existing models (bolded).** Across settings, upgrading the base LLM and vision encoder, as well as pretraining the connector and LLM on sufficient data, consistently benefit fine-grained performance, whereas instruction fine-tuning has a smaller effect. This contrasts with general VQA benchmarks, where the choice of LLM plays a larger role.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

Takeaway 2: Stronger vision encoders (such as DFN-CLIP over CLIP) improve fine-grained classification across VLMs, but only when the encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Across the field, recent VLMs are predominantly trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, involves training on large-scale image-caption datasets using a captioning task to establish alignment between the vision encoder and LLM. The second stage, instruction tuning, refines the VLM with carefully curated multi-round, multi-modal instruction completion data, aligning it with human intent across the broad set of real-world applications VLMs are meant to serve. Here, we explore the impact of these two widely adopted training stages on the fine-grained capabilities of VLMs.

4.3.1 PRETRAINING

Setup. We investigate several critical questions about the pretraining process:

- Is pretraining necessary for LLaVA-architecture models when the base models are already pretrained on billion-scale data?
- How does connector-only pretraining compare with unfreezing the LLM?
- How important is the quality of pretraining data for downstream performance?

To cover a broad spectrum of data quality, we experiment with three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, with high-quality human-annotated detailed captions), and no pretraining. For most experiments, we pretrain only the connector following Liu et al. (2023). In addition, we explore a strategy similar to Deitke et al. (2024), in which we pretrain the connector for 20% of the steps before unfreezing the LLM for the remaining 80%. All configurations are followed by one epoch of finetuning on LLaVA finetuning data.

Findings. Figure 6 illustrates our pretraining ablation results. We observe that connector-only pretraining broadly enhances VLM performance, with the lone exception of the Qwen2 and CLIP ViT-L/14 combination. The benefits are particularly substantial for Vicuna-based VLMs, which show an average improvement of +6.1pp in fine-grained classification and +3.1pp in general VQA performance.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance across settings, while having a more modest effect on general VLM benchmarks.

Perhaps surprisingly, we find minimal differences between pretraining on low-quality web-scraped captions (LLaVA) and highly detailed human annotations (PixMo), despite the wide gap in caption quality between the two. Compared to LLaVA data, PixMo data results in a -1.6pp change in fine-grained performance and a +0.4pp change in general VQA scores.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance, even across markedly different caption sources.

We hypothesize that connector-only pretraining may prevent the model from fully leveraging higher-quality captions. To test this hypothesis, we experiment with pretraining both the LLM and connector, incorporating a 20% connector-only warmup phase. Figure 7 shows that this approach substantially enhances fine-grained classification performance (+5.5pp) without compromising general VQA scores. Importantly, this improvement holds with both LLaVA and PixMo pretraining data, suggesting more generally that training on lower-quality captions does not impair the LLM’s language capabilities.

Takeaway 5: Pretraining both the LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores, regardless of the pretraining caption source.

4.3.2 FINETUNING

Setup. Departing from our previous setup, we begin with the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b) (from which Qwen2-VL-7B-Chat is trained) and finetune on LLaVA-Instruct data (Liu et al., 2023) following our earlier experimental protocol. We compare these results against Qwen2-VL-7B-Chat, which employs 2M in-house examples for instruction finetuning (Wang et al., 2024b). This comparison, which contrasts very different finetuning corpora, helps isolate the effect of the instruction-tuning data.

Findings. Our results indicate that finetuning solely on LLaVA finetuning data slightly reduces both fine-grained and general benchmark performance. Specifically, fine-grained scores decrease by 2.4pp (from 87.9% to 85.5%), while general benchmark scores drop by 1.3pp (from 62.4% to 61.1%). As illustrated in Figure 8, this represents the smallest contribution to fine-grained performance among all of our ablations, suggesting that instruction finetuning plays a less critical role in fine-grained knowledge acquisition than base model selection and pretraining strategy—a ranking we expect to hold broadly.

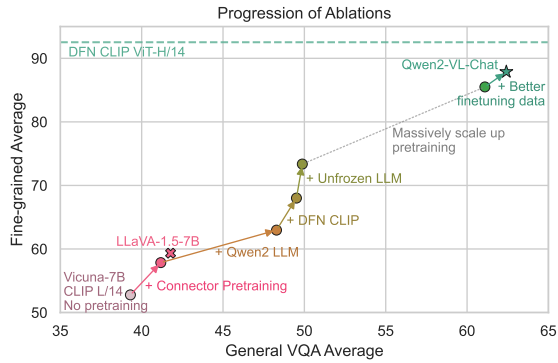


Figure 8: **Progression of combined ablation effects.** Starting from Vicuna-7B and CLIP ViT-L/14, we progressively modify the base LLM, vision encoder, and training settings to raise fine-grained classification accuracy from 52.8% to 73.4%, accounting for more than half of the gap to Qwen2-VL-Chat’s 87.9%. We attribute the remaining difference primarily to Qwen2-VL’s far more extensive pretraining data of 1.4T tokens, pointing to data scale as a broadly important factor.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than the other factors we study.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize our ablation studies, showing how each component contributes to closing the performance gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Beginning with the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021)), we systematically modified the individual components to raise fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. Taken together, this analysis reveals a broadly consistent pattern: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately enhance fine-grained capabilities relative to general VLM abilities, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) accounts for most of the increase in general VQA performance.

Despite these substantial improvements, a notable 12-point gap in fine-grained classification performance remains unexplained by our ablations (Table 1). We consider two broad factors that could account for this discrepancy: architectural differences and pretraining data scale.

While architectural changes could contribute to the difference, our observational results (Section 3.4) and training ablations (Section 4.3) together point strongly to pretraining data scale as the dominant factor. Our experiments with LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) data involved relatively small corpora—each comprising fewer than 1M images and captions, or roughly 200M and 400M tokens, respectively. In contrast, Wang et al. (2024b) report pretraining Qwen2-VL on an extensive 1.4T tokens—orders of magnitude more than in our experiments. This large disparity in pretraining data scale likely accounts for the remaining performance gap, and, more broadly, highlights the critical role of large-scale pretraining in developing VLMs with strong fine-grained classification capabilities.

5 LIMITATIONS & CONCLUSION

Our study has some practical limitations that open up opportunities for future research and that bound the range over which our conclusions have been directly verified. Due to computational constraints, we compared training on <1M data points rather than the billion (B) scale training used by some newer VLMs, leaving open the question of how our observations extend to the largest-scale training regimes. In addition, recent work proposes alternative pretraining strategies, which may interact differently with fine-grained visual understanding and represent a natural direction for broadening the scope of these findings.

540 In this work, we systematically evaluate state-of-the-art vision-language models (VLMs) across
541 fine-grained classification benchmarks spanning multiple visual domains, establishing fine-grained
542 visual classification as a crucial yet broadly underexplored dimension of VLM capability. Through
543 an in-depth analysis of key model components and training paradigms that are common across the
544 VLM landscape, we provide broadly applicable insights into strategies for improving fine-grained
545 classification and, more generally, for enhancing the vision-centric capabilities of VLMs, ultimately
546 strengthening their applicability across the wide range of real-world scenarios that demand precise
547 visual understanding.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	PixMo	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: Detailed results of model ablations across configurations.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: Detailed results of model ablations across the general VQA benchmarks.

A FULL RESULTS

We present the complete ablation results across all configurations—covering aggregated performance, the general VQA benchmarks, and fine-grained classification across its constituent domains—in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: Detailed results of model ablations across the fine-grained classification benchmarks and domains.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As covered in Section 3.2, we evaluate VLMs and base vision encoders on four traditional classification benchmarks that together span a broad set of visual domains:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset covering a wide range of supercategories, each containing multiple fine-grained subcategories, organized according to the WordNet hierarchy. The ImageNet test set contains 50,000 images with classification across 1,000 classes.

However, because of the nature of the ImageNet/WordNet class hierarchy, many classes are too visually distinct from the others to yield hard multiple-choice examples (for instance, only three classes fall under the “person” category). To address this, together with the sheer size of the dataset, we employ human-annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative answer choices can be constructed. Starting from 19,056 annotated examples, we narrow the pool to 5,457 images spanning 928 of the original 1,000 classes. Because this curation leaves the remaining classes unevenly balanced, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species that exemplifies the botanical domain, characterized by high intra-class variation and inter-class similarity. The Flowers dataset consists of 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories representative of the animal domain, with challenging variations in pose, lighting, and occlusion. The Pets dataset consists of 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset comprising 101 food categories that covers the culinary domain, exhibiting substantial visual diversity within each class. The Food dataset consists of 25,250 test examples.

Question generation. For all fine-grained classification test sets other than ImageNet (described above), we generate hard negatives uniformly by performing zero-shot classification on the test examples using an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following

Geigle et al. (2024). We take the top 4 incorrect predictions and shuffle them with the correct label to produce the list of answer choices.

Prompt formatting. In our multiple-choice VQA format, the question always takes the form “What type of object is in this photo?”, a wording deliberately kept general so that it applies across all four domains. The answer choices are formatted as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Since different VLMs are trained with different multiple-choice prompts and formats, we adopt each model’s corresponding format from VLMEvalKit (Duan et al., 2024), for instance: “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use the standard zero-shot evaluation procedure, so there is no question format. Instead, the classnames passed to the text encoder are formatted with the template “a photo of a {classname}”.

B.2 GENERAL VQA

In our initial observational testing from Section 3, we rely directly on the numbers reported on the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard reports the average score across eight diverse VLM benchmarks that together span a wide range of capabilities: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This reported average score is used only in Figure 2 in our paper.

For all further evaluations, both on existing models and on our own models trained in Section 4, we narrow this to a representative subset of multiple-choice VLM benchmarks that enables a more direct and consistent comparison of performance. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples testing a wide range of VLM capabilities spanning perception and reasoning. While most examples deal with coarse-grained perception and visual reasoning tasks, some test “fine-grained perception”—though these categories (such as action recognition, attribute recognition, and OCR) lie in a domain distinct from the fine-grained classification datasets we consider.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset spanning a broad distribution of domains and applications. Though the test set has 10,500 examples, most works report model performance on the validation split, which contains 900 questions. MMMU tests general knowledge tied to visual perception across art, science, engineering, medicine, business, and the humanities.

MMStar (Chen et al., 2024b): A dataset of 1,500 questions selected by human annotators. The examples are chosen to cover a diverse set of tasks, spanning both domains and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). All existing models come with a pre-defined evaluation procedure, and for our own trained models we apply the LLaVA evaluation code, since we exclusively finetune on LLaVA-Instruct data.

C TRAINING

We build on the LLaVA-1.5 training codebase (Liu et al., 2023). For the majority of experiments, we stick to the LLaVA architecture and training procedure.

Pretraining. This means one epoch of pretraining on the LLaVA pretraining data, obtained from CC-3M images and captions, or Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we only use the “caption” field and not the raw audio transcript. During pretraining, we use a batch size of 256. In most of our pretraining runs, we tune only the randomly-initialized MLP connector with a learning rate of 1e-3. When ablating pretraining strategy, we first tune the connector only as warmup, for 20% of steps, then unfreeze the LLM and tune it as well, with a learning rate of 2e-5 for the remaining 80% of steps.

Finetuning. For finetuning, we always train for one epoch on the LLaVA-1.5 finetuning data, which consists of LLaVA-Instruct and VQA training data from various datasets to allow proper benchmarking. During finetuning, we maintain a learning rate of $2e-5$ for the LLM, with a batch size of 128. The vision encoder remains frozen during both phases of training.

For the ablation experiment on the Qwen2-VL architecture, we start with Qwen2-VL-7B Base and train with the same method, unfreezing only the LLM.

Compute. We train all models using 4 nodes of 4 A100 (40GB) GPUs each. Training typically completes within 20 hours, with variation depending on the CLIP model size.

LLaVA vs. PixMo data. In our main text, we refer to the difference in caption quality between LLaVA pretraining data and PixMo pretraining data. LLaVA pretraining data is sourced from CC-3M, which consists of web-scraped captions, commonly known for low quality and relevance. PixMo-Cap is, on the other hand, collected from human annotators recording a 60–90 second audio description of an image, and then combined into a proper image caption (Deitke et al., 2024). A surface analysis of the two datasets shows that the LLaVA data averages 9.8 words per caption, whereas the PixMo data averages 169 words per caption.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”